Compact factors for actual whole crossing charts

14 November 2026; 39 authored Lean statements
Source `95f5d5c917a286103a3c74deaa9db431e5196e81`

Construction. The same completed auxiliary orientation polynomial now has explicit whole-polynomial crossing factors at arbitrary finite precision, for every retained nonzero source node. The proof constructs the factors through actual bivariate coefficients and row classes. No basis, rank witness or favorable coefficient bound is supplied as a premise. All records are unsigned [pcc:unverified].

Actual hypotheses. Let $L = 2^a$, actual ports $x, y < L$, and nonzero integer $t \in \{-L, \dots, L-1\}$ with $v_2(t) = i < u \le a$. Matrices explicitly retain equal-bit-i crossings and zero other entries. Multiplicities m_j are arbitrary naturals. Use the original nonzero negative-pole diagonal c, $w = v_2(c)$, and the same $W_{xy}(T)$, with

$$r = x \bmod 2^u, \quad h = x \operatorname{div} 2^u, \quad \alpha = 2^{a+1}, \quad P_{xy}(Z) = W_{xy}(t + \alpha Z)/c.$$

No nonzero local first jet or positive multiplicity is assumed in the general factor result. The identity origin is unchanged [pcc:unverified].

Actual row polynomial. Put $\delta_r = y - r$, $e_t = 2^u/(t - \delta_r)$, $e_j = 2^u/(2^j - \delta_r)$, $\lambda = \alpha/(t - \delta_r)$, and

$$\mathcal{D}(H, Z) = (1 + \lambda Z + e_t H) \prod_{j \in M_{ry}} (1 + e_j H)^{m_j}, \qquad N(Z) = (1 + \lambda Z) P_{ry}(Z).$$

Actual masks and denominators give even e_t, e_j, λ. Substituting the actual integer h gives exactly the previous component's actual denominator $D(Z)$, with $DP_{xy} = N$ and $D - 1$ of content one. These bridges are derived from the original completed polynomials, not assumed [pcc:unverified].

Define

$$\mathcal{A}(H, Z) = N(Z) \sum_{n<k} (1 - \mathcal{D}(H, Z))^n.$$

Its coefficient of H^b has polynomial content at least b in Z. The proof derives this using actual even slopes and explicit polynomial closure. At integer h, removing all terms $b \ge k$ has content-k error. The remaining finite sum approximates every coefficient of P_{xy} at precision k [pcc:unverified].

Literal factors and actual weighted residual

For each representative $r \in \mathrm{Fin}(2^u)$ and $b \in \mathrm{Fin}(k)$, define polynomial matrices

$$U_{x,(r,b)} = \begin{cases} 2^b h^b & r = x \bmod 2^u, \\ 0 & \text{otherwise,} \end{cases} \qquad V_{(r,b),y} = 2^{-b}[H^b]\mathcal{A}_{ry}(H, Z),$$

where the right factor is explicitly zero if r, y do not cross at bit i. The left scalar is viewed as a constant polynomial in Z. The representative construction depends only on the row class, so finite matrix multiplication selects its actual truncated row polynomial [pcc:unverified].

All right factors are integral in every Z coefficient because $[H^b]\mathcal{A}$ has content at least b. The left has content at least b. For the normalized crossing polynomial matrix P^{cross},

$$\mathrm{Content}_k(P^{\mathrm{cross}}_{xy} - (UV)_{xy}).$$

The factors are constructed for the whole polynomial, including higher chart coefficients, rather than just its first jet [pcc:unverified].

Scale the left factor by the same actual c and call it $\widehat{U}$. For the actual unnormalized crossing chart matrix W^{cross},

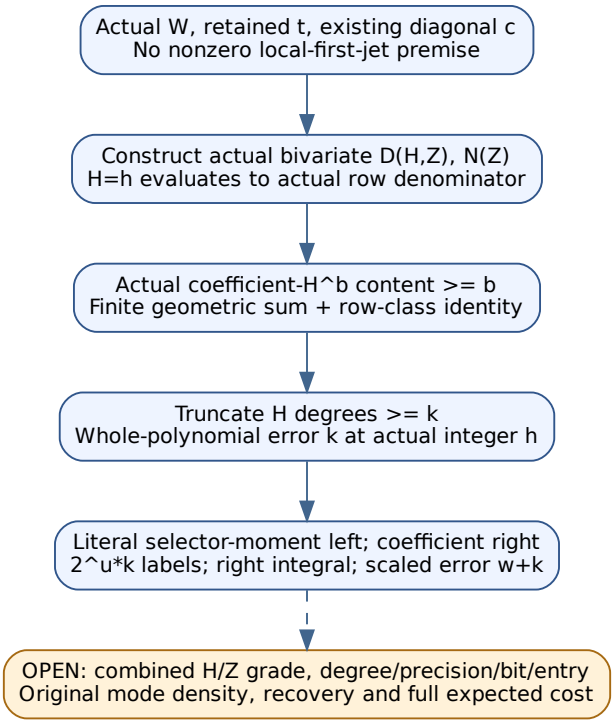
$$\mathrm{Content}_{w+b}(\widehat{U}_{x,(r,b)}), \qquad \mathrm{Content}_0(V_{(r,b),y}), \qquad \mathrm{Content}_{w+k}(W^{\mathrm{cross}}_{xy} - (\widehat{U}V)_{xy}).$$

Scaled-left integrality follows when $w \geq 0$. Exact inner-label cardinality is $2^u k$, including the empty $k = 0$ case. Every label is charged [pcc:unverified].

Scope at anchor zeros. The derivation does not cancel a local first jet. Positive anchors where that jet vanishes remain included. Previous first-jet zero controls concern first jets only. This component makes no low-rank statement for all unmasked W or for its identity origin [pcc:unverified].

What the count does not price. The component gives whole-polynomial content and explicit factors. It does not yet bound each right factor's necessary degree at precision k, coefficient bit sizes, entry construction, creation precision or total running time. Those are the next proof tasks.

Dependency map and original mode costs



The reviewed original interface allows at most $6M/D_i$ modes per weight, $D_i = L/2^i$, and at most $3M/L$ packets. The present construction retains 2^u row classes at every moment index; that can exceed the old 2^{i+1} tile inventory. A factor count alone does not establish a runtime gain [pcc:unverified].

Next discriminating proof. Establish combined H/Z coefficient grading, then truncate right Z degree at the requested residual while retaining the actual matrix error. Price all coefficient construction, entry and bit operations, degree, precision, recovery and recursion against the original weighted mode interface. Actual elliptic relation, efficient public finder, field and scalar extraction, retries and full expected bit cost remain explicit obligations. No sub-exponential runtime or algorithm conclusion follows from this component.

Source bindings, checks and preservation

The isolated project builds 39 authored statements without warnings. The pinned shared audit covers39 declarations,0 extra trusted axioms, exit0. Import-inclusive fresh replay completed exit0/exact42-byte stdout. Five central printed dependencies contain only `propext`, `Classical.choice`, `Quot.sound`. All records are unsigned [pcc:unverified].

`denom_rows_eval`	PCC-20261114-4f155f
`actual_low_rows_precision`	PCC-20261114-e87981
`poly_right_integral`	PCC-20261114-8af989
`actual_compact_poly_precision`	PCC-20261114-ce6456
`actual_compact_weighted_profile`	PCC-20261114-6a7512

Exact statements govern every hypothesis. Source/configuration pins, full statement bindings and the prospective contract are in

`../evidence/source-freeze.json`
`../evidence/claim-pins.json`
`../PROTOCOL.md`.

Gate data is PCC-20261114-92f882; fresh data PCC-20261114-256a8f. Local accepted008 data is PCC-20261114-91d645; it alone is not independent gate/replay coverage [pcc:unverified].

Pinned environment. Lean4.35.0, CryptoLean and Mathlib revisions:

`148f5dc389b341401e06d76abe7783c4ccf42f2e`
`cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`.

The source is byte-identical accepted008. All eight local drafts and raw build logs/exits remain in the parent bivariate evidence directory. Failed attempts include missing Content/Jet namespaces, an ambiguous slope, and a misidentified valuation-lemma namespace. Failed elaboration diagnostics including `sorryAx` remain raw evidence; accepted008 contains no such dependency. No linter was disabled or favorable hypothesis added.

Sources checked. KB unavailable as journaled. Read exact Higher, Node, Moment and Content coefficient, evaluation, row-class and truncation APIs; pinned Mathlib polynomial and power-series truncation APIs; primary D.4.3–7 and prior contraction cost/precision reviews. Standard bivariate polynomials and geometric sums are reused, with no novelty, priority, fastest-method or measured timing claim.

No native crypto, attack, F4/msolve, timing, scaling, paid or cloud experiment ran. The original norm fixture remains unlaunched under its unchanged transport blocker. Canonical IC graphs/leaderboards have no new measured result; this diagram describes proof dependencies.

Higher83 and Node90 checkpoint artifacts and all earlier parent sources, reports, claims and raw outputs remain exact. Protected concurrent PCC histories still need legitimate canonical-maintainer reconciliation; auto-merge stays off. No choose-side, rechain, claim rewrite, signature, trust change, prefix waiver, force push or merge is attempted.